

Chronic Digoxin Toxicity



Section I: Scenario Demographics

Scenario Title:	Chronic Digoxin Toxicity
Date of Development:	26/04/2018
Target Learning Group:	☐ Juniors (PGY 1 - 2) ☒ Seniors (PGY $\geq$ 3) ☐ All Groups

Section II: Scenario Developers

Scenario Developer(s):	Dr. Rebecca Shaw, Dr. Nemat Alsaba, Professor Victoria Brazil
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Section III: Curriculum Integration

Learning Goals & Objectives	
Educational Goal:	To practice team-based assessment of an elderly patient presenting with a fall and management of digoxin toxicity.
CRM Objectives:	- Lead the team effectively by prioritizing tasks and clearly communicating decisions - Communicate clearly with patient and update patient on management
Medical Objectives:	- Initiate appropriate work-up taking into account broad differential diagnosis for falls in the elderly - Recognize and treat hyperkalemia as a possible cause of bradycardia - Begin appropriate resuscitation measures for an unstable bradycardic patient - Identify and treat chronic digoxin toxicity with specific therapies (Digifab)

Case Summary: Brief Summary of Case Progression and Major Events

An 85-year-old man presents after a fall at home. He is complaining of dizziness and has a HR of 30. Further assessment reveals chronic digoxin toxicity and a concurrent UTI with acute renal failure. The patient requires management of his bradycardia and acute renal failure with specific management of chronic digoxin toxicity including a discussion with toxicology and administration of Digibind.

References

<https://lifeinthefastlane.com/tox-library/toxicant/anti-arrhythmics/digoxin/>



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Section IV: Scenario Script

A. Clinical Vignette: To Read Aloud at Beginning of Case			
Bertie is an 85-year-old man who has been brought in after a fall at home. He says he is feeling dizzy and has a HR of 30 on the monitor. I haven't had much of a chance to take more of a history from him but he has a list of medications with him and seems ok from the fall other than a bruise on his head.			
B. Scenario Cast & Realism			
Patient:	☐ Computerized Mannequin	Realism:	☒ Conceptual
	☐ Mannequin		☐ Physical
	☒ Standardized Patient		☐ Emotional/Experiential
	☐ Hybrid		☐ Other:
	☐ Task Trainer		☐ N/A
		<i>Select most important dimension(s)</i>	
Confederates	Brief Description of Role		
ED RN	Provides initial handover. To assist at the bedside.		
Patient (SP)	You came to hospital as you had a fall at home today. You don't really remember the fall itself but you remember that you were walking towards the kitchen to make a cup of tea and the next thing you remember is waking up on the floor. You have a mild headache after the fall but otherwise no pain and no other injuries. You had no chest pain, shortness of breath or palpitations before or after the fall. However, you are feeling very dizzy at the moment and you were feeling dizzy before you fell. You have been feeling very nauseated for the last week and had a vomit yesterday. You went to see the GP a couple of days ago because you had been having difficulty passing urine and had some mild lower "tummy pain". Your stool has been more loose than usual for the last week. The GP started you on some antibiotics to treat a urine infection. You have been feeling very thirsty for the last couple of days and a bit more muddled than usual finding it difficult to remember things. You also noticed this morning that your vision seems slightly more blurred than usual. Your other medical problems are irregular heartbeat, a previous heart attack 5 years ago, COPD. You are usually on aspirin, a fluid tablet and heart medication (you hand the doctor a medication list that you keep in your wallet). You have no allergies. You live at home independently with your wife. You used to smoke but quit 10 years ago and you have a glass of wine a night. You are concerned about your current situation. If you have not been given any update on your situation ask couple of questions to prompt them: can I ask what is happening? will I be alright? when I can go home?		
Toxicologist	To take the referral from the participates and give advice /plan as needed.		



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C. Required Monitors

☒ EKG Leads/Wires	☐ Temperature Probe	☐ Central Venous Line
☒ NIBP Cuff	☒ Defibrillator Pads	☐ Capnography
☒ Pulse Oximeter	☐ Arterial Line	☐ Other:

D. Required Equipment

☒ Gloves	☒ Nasal Prongs	☐ Scalpel
☒ Stethoscope	☒ Venturi Mask	☐ Tube Thoracostomy Kit
☒ Defibrillator	☒ Non-Rebreather Mask	☐ Cricothyroidotomy Kit
☒ IV Bags/Lines	☒ Bag Valve Mask	☐ Thoracotomy Kit
☒ IV Push Medications	☒ Laryngoscope	☐ Central Line Kit
☐ PO Tabs	☐ Video Assisted Laryngoscope	☐ Arterial Line Kit
☐ Blood Products	☒ ET Tubes	☐ Other:
☐ Intraosseous Set-up	☐ LMA	☐ Other:

E. Moulage

Abrasion to right forehead

F. Approximate Timing

Set-Up:	5 min	Scenario:	20 min	Debriefing:	20 min
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Section V: Patient Data and Baseline State

A. Patient Profile and History					
Patient Name: Bertie Banks		Age: 85		Weight: 78 kg	
Gender: ☒ M ☐ F		Code Status: presumed full			
Chief Complaint: Fall at home					
History of Presenting Illness: The patient fell walking to the kitchen after feeling dizzy and woke up on the floor with a mild headache. There are otherwise no injuries or other pre-syncopal symptoms. The patient has been complaining of nausea for the last week with a vomit yesterday. The GP recently commenced antibiotics for a UTI due to increased urinary frequency. The patient also has loose stool for a week and slightly blurred vision today.					
Past Medical History:	AF		Medications:	Digoxin	
	Congestive Heart Failure			Amiodarone	
	Previous MI			Metoprolol	
	COPD			Furosemide	
				Spironolactone	
				Aspirin	
Allergies: Nil					
Social History: Lives at home independently with wife. Ex-smoker. Drinks a glass of wine a night.					
Family History: Nil of note					
Review of Systems:	CNS:	Mild headache			
	HEENT:	Blurry vision			
	CVS:	Intermittent dizziness for a week. Currently feels light headed			
	RESP:	Nil			
	GI:	Loose stool and mild lower abdominal pain for a week			
	GU:	Increased urinary frequency			
	MSK:	nil	INT:	nil	
B. Baseline Simulator State and Physical Exam					
☐ No Monitor Display		☒ Monitor On, no data displayed		☐ Monitor on Standard Display	
HR: 30/min	BP: 130/70	RR: 16/min	O ₂ SAT: 98% RA		
Rhythm: AF	T: 36.8°C	Glucose: 5.2 mmol/L	GCS: 15 (E 4 V 5 M 6)		
General Status: Looks comfortable in bed, abrasion to right forehead, looks dry					
CNS:	Mild headache, moving all 4 limbs normally, normal neuro exam				
HEENT:	Nil of note				
CVS:	Irregularly irregular heart sounds				
RESP:	Chest clear				
ABDO:	Soft, very mild suprapubic tenderness, nil signs of peritonism, BS present				
GU:	Increased urinary frequency				
MSK:	Nil	SKIN:	nil		



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Section VI: Scenario Progression

Scenario States, Modifiers and Triggers			
Patient State	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State	
1. Baseline State Rhythm: AF HR: 30/min BP: 130/70 RR: 16/min O ₂ SAT: 98% RA T: 36.8°C	Looks comfortable in bed	<u>Learner Actions</u> ☐ History and examination ☐ IV access ☐ Bloods: VBG, lytes, extended lytes, digoxin level, consider troponin ☐ Urine R+M, C+S ☐ ECG ☐ Place defibrillator pads ☐ ± Empiric IV antibiotics	<u>Modifiers</u> <i>Changes to patient condition based on learner action</i> - Pacing attempted → no capture <u>Triggers</u> <i>For progression to next state</i> - All actions complete or after 5 mins → 2. Deterioration
2. Deterioration Rhythm: AF HR: 30/min BP: 95/50 RR: 22/min O ₂ SAT: 95% RA	Remains talkative and GCS 15 throughout Critical VBG result (including potassium) given at onset of state	<u>Learner Actions</u> ☐ CXR ☐ Fast scan ☐ IV fluid ☐ Atropine ☐ IV fluid bolus ☐ ± Epinephrine infusion at bedside ☐ Calcium gluconate 1g iv administered SLOWLY ☐ Insulin/dextrose	<u>Modifiers</u> - HR to remain 30/min if atropine given - Pacing attempted → no capture <u>Triggers</u> - All actions complete or after 7 minutes → 3. Hypotension
3. Hypotension Rhythm: AF HR: 30/min BP: 80/40 RR: 22/min O ₂ SAT: 95% RA	Slower to respond but still alert Digoxin result given at onset of state	<u>Learner Actions</u> ☐ Discussion with toxicology ☐ Administer Digifab: 3 vials (calculated dose) or 5 vials (empiric dose) ☐ ± Start epinephrine or isoproterenol infusion ☐ ± Transcutaneous pacing	<u>Modifiers</u> - Epi infusion → BP 85/50, HR 35 - Pacing attempted → no capture <u>Triggers</u> - Digifab given → 4. Resolution
4. Resolution Rhythm: AF HR: 50/min BP: 135/70 RR: 20/min O ₂ SAT: 96% RA	Patient starts to feel better	<u>Learner Actions</u> ☐ Consult admitting service ☐ CT head ☐ ECG ☐ Repeat potassium	END CASE PRN



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Section VII: Supporting Documents, Laboratory Results, & Multimedia

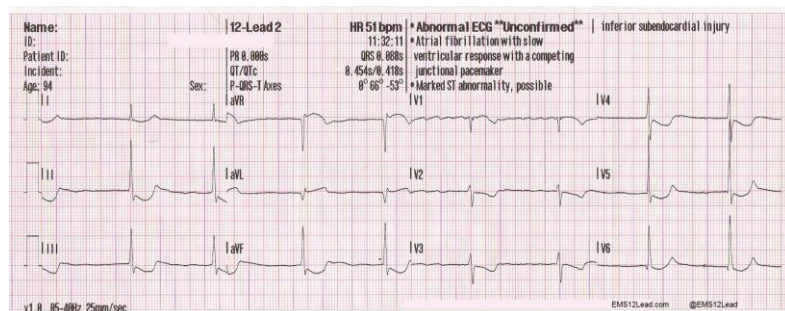
Laboratory Results

Laboratory Results						
Na: 135	K: 6.9	Cl: 110	HCO ₃ : 23	BUN: 12.7	Cr: 190	Glu: 5.2
Ca: 1.2		Mg: 0.8		PO ₄ :		Albumin: 31
VBG	pH: 7.34	PCO ₂ : 55	PO ₂ : 94	HCO ₃ : 21	Lactate:2.1	
WBC: 14.0		Hg: 130		Hct: 0.35		Plt: 175
Digoxin level: 3.8 ng/ml (0.2-0.8)						

Images (ECGs, CXRs, etc.)

ECG:

(ECG source: wp-content/uploads/sites/42/2014/01/digitalis_ECG)



CXR:



(CXR source:

<https://radiopaedia.org/images/220869>)



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Section VIII: Debriefing Guide

General Debriefing Plan	
☐ Individual	☒ Group
☐ With Video	☒ Without Video
Objectives	
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Sample Questions for Debriefing	
Medical What differentials did you consider as potential causes for this patient fall? What are the challenges in dealing with Geriatric Toxicology? Have you considered another concurrent drug toxicity (beta blocker)? What are the features of chronic digoxin toxicity? How was this case different? How do you explain the hyperkalemia in this case? What is the role of calcium administration in hyperkalemia due to digoxin toxicity? What is the possible harm associated with giving calcium? How did you manage this hemodynamically unstable patient's bradycardia? What is the management of chronic digoxin toxicity?	
CRM What were your priorities in the management of this patient? Was closed-loop communication used in this high-stress scenario? What strategies did the team utilize to effectively achieve these management priorities?	
Key Moments	
Team based assessment of patient presenting after a fall	
Recognition of chronic digoxin toxicity as cause of fall	
Management of chronic digoxin toxicity and discussion with toxicologist	

